

Chapter 11

Principal Components Analysis

Chapter 10 presented the basic geometric tools needed to produce a lower dimensional description of the rows and columns of a multivariate data matrix. Principal components analysis (PCA) has the same objective with the exception that the rows of the data matrix $\mathcal{X}$ will now be considered as observations from a p -variate random variable X . The principle idea of reducing the dimension of X is achieved through linear combinations. Low dimensional linear combinations are often easier to interpret and serve as an intermediate step in a more complex data analysis. More precisely one looks for linear combinations which create the largest spread among the values of X . In other words, one is searching for linear combinations with the largest variances.

Section 11.1 introduces the basic ideas and technical elements behind principal components. No particular assumption will be made on X except that the mean vector and the covariance matrix exist. When reference is made to a data matrix $\mathcal{X}$ in Sect. 11.2, the empirical mean and covariance matrix will be used. Section 11.3 shows how to interpret the principal components by studying their correlations with the original components of X . Often analyses are performed in practice by looking at two-dimensional scatterplots. Section 11.4 develops inference techniques on principal components. This is particularly helpful in establishing the appropriate dimension reduction and thus in determining the quality of the resulting lower dimensional representations. Since principal component analysis is performed on covariance matrices, it is not scale invariant. Often, the measurement units of the components of X are quite different, so it is reasonable to standardise the measurement units. The normalised version of principal components is defined in Sect. 11.5. In Sect. 11.6 it is discovered that the empirical principal components are the factors of appropriate transformations of the data matrix. The classical way of defining principal components through linear combinations with respect to the largest variance is described here in geometric terms, i.e. in terms of the optimal fit within subspaces generated by the columns and/or the rows of $\mathcal{X}$ as was discussed in Chap. 10. Section 11.9 concludes with additional examples.

11.1 Standardised Linear Combination

The main objective of PCA is to reduce the dimension of the observations. The simplest way of dimension reduction is to take just one element of the observed vector and to discard all others. This is not a very reasonable approach, as we have seen in the earlier chapters, since strength may be lost in interpreting the data. In the bank notes example we have seen that just one variable (e.g. $X_1 = \text{length}$) had no discriminatory power in distinguishing counterfeit from genuine bank notes. An alternative method is to weight all variables equally, i.e. to consider the simple average $p^{-1} \sum_{j=1}^p X_j$ of all the elements in the vector $X = (X_1, \dots, X_p)^\top$. This again is undesirable, since all of the elements of X are considered with equal importance (weight).

A more flexible approach is to study a weighted average, namely

$$\delta^\top X = \sum_{j=1}^p \delta_j X_j, \quad \text{such that} \quad \sum_{j=1}^p \delta_j^2 = 1. \quad (11.1)$$

The weighting vector $\delta = (\delta_1, \dots, \delta_p)^\top$ can then be optimised to investigate and to detect specific features. We call (11.1) a standardised linear combination (SLC). Which SLC should we choose? One aim is to maximise the variance of the projection $\delta^\top X$, i.e. to choose δ according to

$$\max_{\{\delta: \|\delta\|=1\}} \text{Var}(\delta^\top X) = \max_{\{\delta: \|\delta\|=1\}} \delta^\top \text{Var}(X) \delta. \quad (11.2)$$

The interesting “directions” of δ are found through the spectral decomposition of the covariance matrix. Indeed, from Theorem 2.5, the direction δ is given by the eigenvector γ_1 corresponding to the largest eigenvalue λ_1 of the covariance matrix $\Sigma = \text{Var}(X)$.

Figures 11.1 and 11.2 show two such projections (SLCs) of the same data set with zero mean. In Fig. 11.1 an arbitrary projection is displayed. The upper window shows the data point cloud and the line onto which the data are projected. The middle window shows the projected values in the selected direction. The lower window shows the variance of the actual projection and the percentage of the total variance that is explained.

Figure 11.2 shows the projection that captures the majority of the variance in the data. This direction is of interest and is located along the main direction of the point cloud. The same line of thought can be applied to all data orthogonal to this direction leading to the second eigenvector. The SLC with the highest variance obtained from maximising (11.2) is the first principal component (PC) $y_1 = \gamma_1^\top X$. Orthogonal to the direction γ_1 we find the SLC with the second highest variance: $y_2 = \gamma_2^\top X$, the second PC.

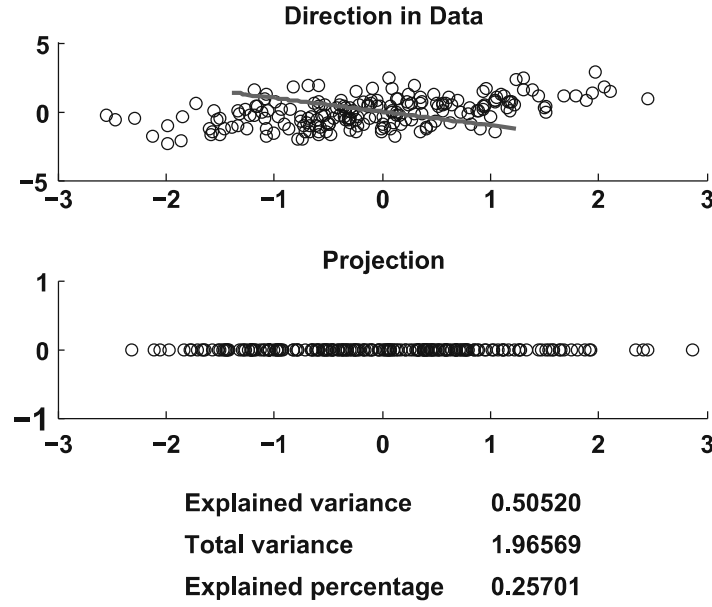


Fig. 11.1 An arbitrary SLC  MVApcasimu

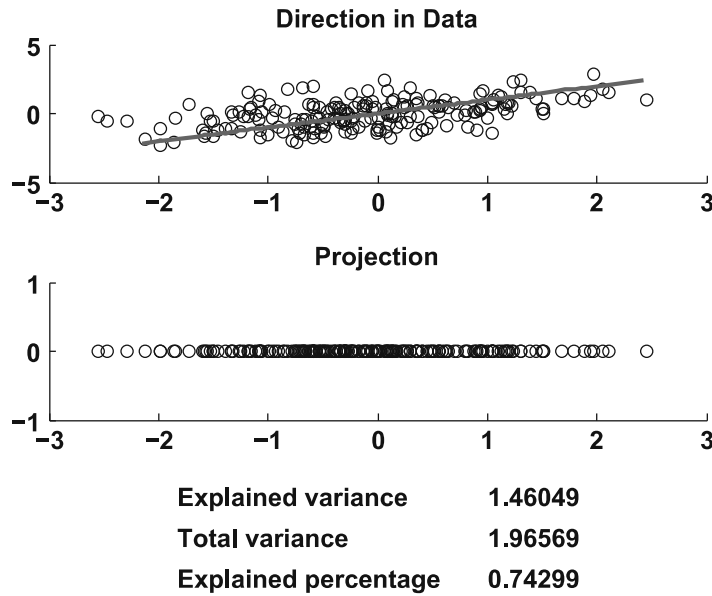


Fig. 11.2 The most interesting SLC  MVApcasimu

Proceeding in this way and writing in matrix notation, the result for a random variable X with $\mathbf{E}(X) = \mu$ and $\mathbf{Var}(X) = \Sigma = \Gamma\Lambda\Gamma^\top$ is the PC transformation which is defined as

$$Y = \Gamma^\top(X - \mu). \quad (11.3)$$

Here we have centred the variable X in order to obtain a zero mean PC variable Y .

Example 11.1 Consider a bivariate normal distribution $N(0, \Sigma)$ with $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ and $\rho > 0$ (see Example 3.13). Recall that the eigenvalues of this matrix are $\lambda_1 = 1 + \rho$ and $\lambda_2 = 1 - \rho$ with corresponding eigenvectors

$$\gamma_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \gamma_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The PC transformation is thus

$$Y = \Gamma^\top (X - \mu) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} X$$

or

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} X_1 + X_2 \\ X_1 - X_2 \end{pmatrix}.$$

So the first principal component is

$$Y_1 = \frac{1}{\sqrt{2}}(X_1 + X_2)$$

and the second is

$$Y_2 = \frac{1}{\sqrt{2}}(X_1 - X_2).$$

Let us compute the variances of these PCs using formulas (4.22)–(4.26):

$$\begin{aligned} \text{Var}(Y_1) &= \text{Var} \left\{ \frac{1}{\sqrt{2}}(X_1 + X_2) \right\} = \frac{1}{2} \text{Var}(X_1 + X_2) \\ &= \frac{1}{2} \{ \text{Var}(X_1) + \text{Var}(X_2) + 2 \text{Cov}(X_1, X_2) \} \\ &= \frac{1}{2} (1 + 1 + 2\rho) = 1 + \rho \\ &= \lambda_1. \end{aligned}$$

Similarly we find that

$$\text{Var}(Y_2) = \lambda_2.$$

This can be expressed more generally and is given in the next theorem.

Theorem 11.1 For a given $X \sim (\mu, \Sigma)$ let $Y = \Gamma^\top(X - \mu)$ be the PC transformation. Then

$$\mathbb{E} Y_j = 0, \quad j = 1, \dots, p \quad (11.4)$$

$$\text{Var}(Y_j) = \lambda_j, \quad j = 1, \dots, p \quad (11.5)$$

$$\text{Cov}(Y_i, Y_j) = 0, \quad i \neq j \quad (11.6)$$

$$\text{Var}(Y_1) \geq \text{Var}(Y_2) \geq \dots \geq \text{Var}(Y_p) \geq 0 \quad (11.7)$$

$$\sum_{j=1}^p \text{Var}(Y_j) = \text{tr}(\Sigma) \quad (11.8)$$

$$\prod_{j=1}^p \text{Var}(Y_j) = |\Sigma|. \quad (11.9)$$

Proof To prove (11.6), we use γ_i to denote the i th column of Γ . Then

$$\text{Cov}(Y_i, Y_j) = \gamma_i^\top \text{Var}(X - \mu) \gamma_j = \gamma_i^\top \text{Var}(X) \gamma_j.$$

As $\text{Var}(X) = \Sigma = \Gamma \Lambda \Gamma^\top$, $\Gamma^\top \Gamma = \mathcal{I}$, we obtain via the orthogonality of Γ :

$$\gamma_i^\top \Gamma \Lambda \Gamma^\top \gamma_j = \begin{cases} 0 & i \neq j, \\ \lambda_i & i = j. \end{cases}$$

In fact, as $Y_i = \gamma_i^\top (X - \mu)$ lies in the eigenvector space corresponding to γ_i , and eigenvector spaces corresponding to different eigenvalues are orthogonal to each other, we can directly see Y_i and Y_j are orthogonal to each other, so their covariance is 0. $\square$

The connection between the PC transformation and the search for the best SLC is made in the following theorem, which follows directly from (11.2) and Theorem 2.5.

Theorem 11.2 There exists no SLC that has larger variance than $\lambda_1 = \text{Var}(Y_1)$.

Theorem 11.3 If $Y = a^\top X$ is an SLC that is not correlated with the first k PCs of X , then the variance of Y is maximised by choosing it to be the $(k + 1)$ -st PC.



Summary

$\hookrightarrow$ An SLC is a weighted average $\delta^\top X = \sum_{j=1}^p \delta_j X_j$ where δ is a vector of length 1.

Summary (continued)	
$\hookrightarrow$	Maximising the variance of $\delta^\top X$ leads to the choice $\delta = \gamma_1$, the eigenvector corresponding to the largest eigenvalue λ_1 of $\Sigma = \text{Var}(X)$. This is a projection of X into the one-dimensional space, where the components of X are weighted by the elements of γ_1 . $Y_1 = \gamma_1^\top (X - \mu)$ is called the first principal component (PC).
$\hookrightarrow$	This projection can be generalised for higher dimensions. The PC transformation is the linear transformation $Y = \Gamma^\top (X - \mu)$, where $\Sigma = \text{Var}(X) = \Gamma \Lambda \Gamma^\top$ and $\mu = \mathbf{E} X$. $Y_1, Y_2, \dots, Y_p$ are called the first, second, ..., and p -th PCs.
$\hookrightarrow$	The PCs have zero means, variance $\text{Var}(Y_j) = \lambda_j$, and zero covariances. From $\lambda_1 \geq \dots \geq \lambda_p$ it follows that $\text{Var}(Y_1) \geq \dots \geq \text{Var}(Y_p)$. It holds that $\sum_{j=1}^p \text{Var}(Y_j) = \text{tr}(\Sigma)$ and $\prod_{j=1}^p \text{Var}(Y_j) = \Sigma $.
$\hookrightarrow$	If $Y = a^\top X$ is an SLC which is not correlated with the first k PCs of X , then the variance of Y is maximised by choosing it to be the $(k + 1)$ -st PC.

11.2 Principal Components in Practice

In practice the PC transformation has to be replaced by the respective estimators: μ becomes $\bar{x}$, Σ is replaced by $\mathcal{S}$, etc. If g_1 denotes the first eigenvector of $\mathcal{S}$, the first principal component is given by $y_1 = (\mathcal{X} - \mathbf{1}_n \bar{x}^\top) g_1$. More generally if $\mathcal{S} = \mathcal{G} \mathcal{L} \mathcal{G}^\top$ is the spectral decomposition of $\mathcal{S}$, then the PCs are obtained by

$$\mathcal{Y} = (\mathcal{X} - \mathbf{1}_n \bar{x}^\top) \mathcal{G}. \quad (11.10)$$

Note that with the centering matrix $\mathcal{H} = \mathcal{I} - (n^{-1} \mathbf{1}_n \mathbf{1}_n^\top)$ and $\mathcal{H} \mathbf{1}_n \bar{x}^\top = 0$ we can write

$$\begin{aligned} \mathcal{S}_Y &= n^{-1} \mathcal{Y}^\top \mathcal{H} \mathcal{Y} = n^{-1} \mathcal{G}^\top (\mathcal{X} - \mathbf{1}_n \bar{x}^\top)^\top \mathcal{H} (\mathcal{X} - \mathbf{1}_n \bar{x}^\top) \mathcal{G} \\ &= n^{-1} \mathcal{G}^\top \mathcal{X}^\top \mathcal{H} \mathcal{X} \mathcal{G} = \mathcal{G}^\top \mathcal{S} \mathcal{G} = \mathcal{L} \end{aligned} \quad (11.11)$$

where $\mathcal{L} = \text{diag}(\ell_1, \dots, \ell_p)$ is the matrix of eigenvalues of $\mathcal{S}$. Hence the variance of y_i equals the eigenvalue ℓ_i !

The PC technique is sensitive to scale changes. If we multiply one variable by a scalar we obtain different eigenvalues and eigenvectors. This is due to the fact that

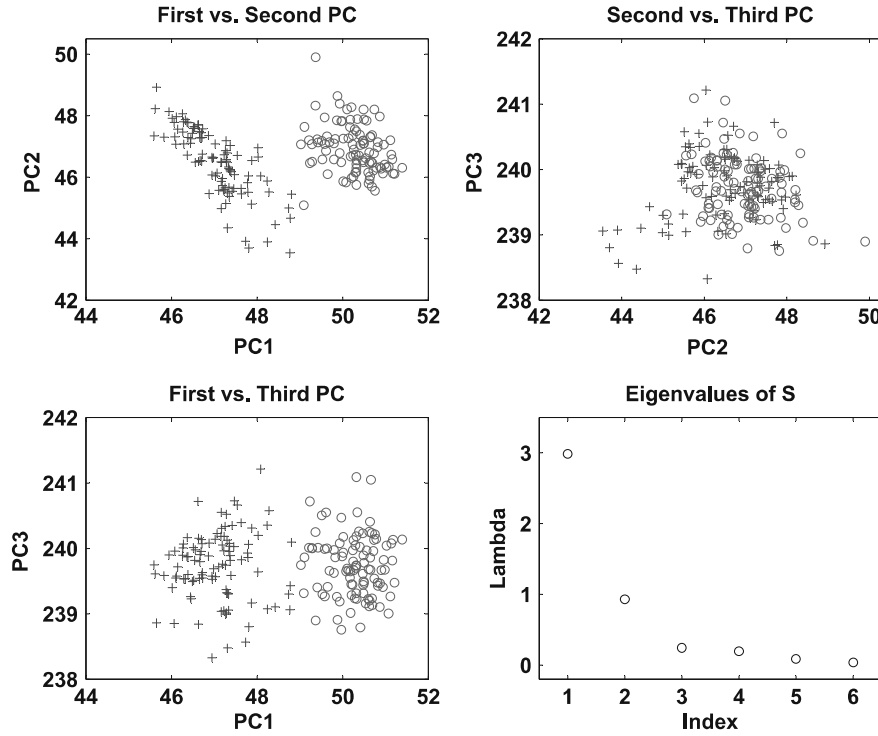


Fig. 11.3 Principal components of the bank data  MVApcabank

an eigenvalue decomposition is performed on the covariance matrix and not on the correlation matrix (see Sect. 11.5). The following warning is therefore important:



The PC transformation should be applied to data that have approximately the same scale in each variable.

Example 11.2 Let us apply this technique to the bank data set. In this example we do not standardise the data. Figure 11.3 shows some PC plots of the bank data set. The genuine and counterfeit bank notes are marked by “o” and “+”, respectively.

Recall that the mean vector of $\mathcal{X}$ is

$$\bar{x} = (214.9, 130.1, 129.9, 9.4, 10.6, 140.5)^T.$$

The vector of eigenvalues of $\mathcal{S}$ is

$$\ell = (2.985, 0.931, 0.242, 0.194, 0.085, 0.035)^T.$$

The eigenvectors g_j are given by the columns of the matrix

$$\mathcal{G} = \begin{pmatrix} -0.044 & 0.011 & 0.326 & 0.562 & -0.753 & 0.098 \\ 0.112 & 0.071 & 0.259 & 0.455 & 0.347 & -0.767 \\ 0.139 & 0.066 & 0.345 & 0.415 & 0.535 & 0.632 \\ 0.768 & -0.563 & 0.218 & -0.186 & -0.100 & -0.022 \\ 0.202 & 0.659 & 0.557 & -0.451 & -0.102 & -0.035 \\ -0.579 & -0.489 & 0.592 & -0.258 & 0.085 & -0.046 \end{pmatrix}.$$

The first column of $\mathcal{G}$ is the first eigenvector and gives the weights used in the linear combination of the original data in the first PC.

Example 11.3 To see how sensitive the PCs are to a change in the scale of the variables, assume that X_1, X_2, X_3 and X_6 are measured in cm and that X_4 and X_5 remain in mm in the bank data set. This leads to:

$$\bar{x} = (21.49, 13.01, 12.99, 9.41, 10.65, 14.05)^\top.$$

The covariance matrix can be obtained from S in (3.4) by dividing rows 1, 2, 3, 6 and columns 1, 2, 3, 6 by 10. We obtain:

$$\ell = (2.101, 0.623, 0.005, 0.002, 0.001, 0.0004)^\top$$

which clearly differs from Example 11.2. Only the first two eigenvectors are given:

$$g_1 = (-0.005, 0.011, 0.014, 0.992, 0.113, -0.052)^\top$$

$$g_2 = (-0.001, 0.013, 0.016, -0.117, 0.991, -0.069)^\top.$$

Comparing these results to the first two columns of $\mathcal{G}$ from Example 11.2, a completely different story is revealed. Here the first component is dominated by X_4 (lower margin) and the second by X_5 (upper margin), while all of the other variables have much less weight. The results are shown in Fig. 11.4. Section 11.5 will show how to select a reasonable standardisation of the variables when the scales are too different.



Summary

↪ The scale of the variables should be roughly the same for PC transformations.

Summary (continued)	
↪	For the practical implementation of PCA we replace μ by the mean $\bar{x}$ and Σ by the empirical covariance $\mathcal{S}$. Then we compute the eigenvalues $\ell_1, \dots, \ell_p$ and the eigenvectors $g_1, \dots, g_p$ of $\mathcal{S}$. The graphical representation of the PCs is obtained by plotting the first PC vs. the second (and eventually vs. the third).
↪	The components of the eigenvectors g_i are the weights of the original variables in the PCs.

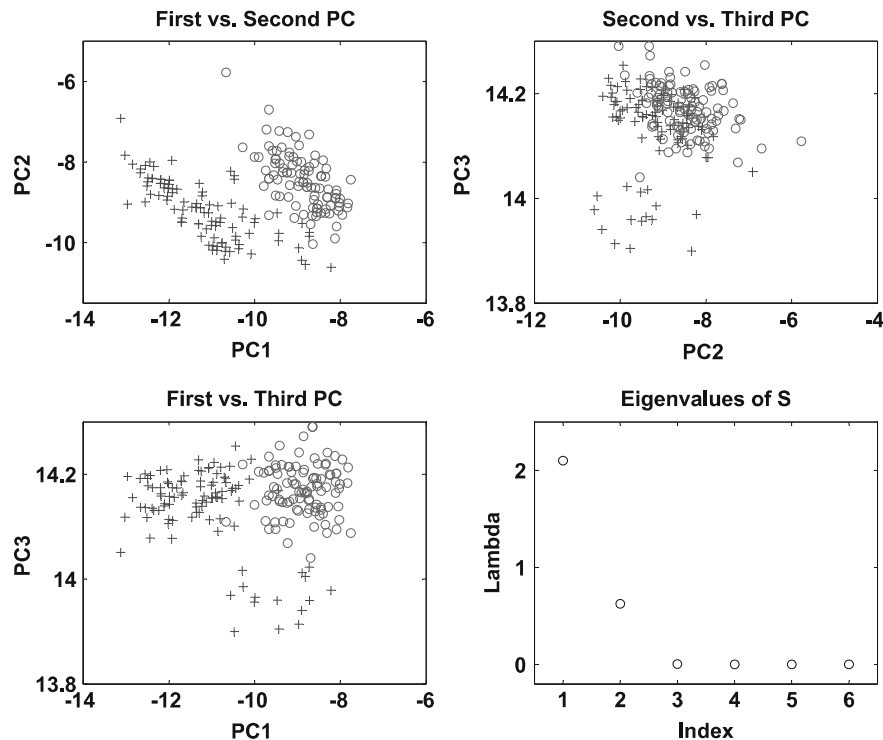


Fig. 11.4 Principal components of the rescaled bank data  MVApcabankr

11.3 Interpretation of the PCs

Recall that the main idea of PC transformations is to find the most informative projections that maximise variances. The most informative SLC is given by the first eigenvector. In Sect. 11.2 the eigenvectors were calculated for the bank data. In particular, with centred x 's, we had:

$$\begin{aligned} y_1 &= -0.044x_1 + 0.112x_2 + 0.139x_3 + 0.768x_4 + 0.202x_5 - 0.579x_6 \\ y_2 &= 0.011x_1 + 0.071x_2 + 0.066x_3 - 0.563x_4 + 0.659x_5 - 0.489x_6 \end{aligned}$$

and

$x_1 = \text{length}$
 $x_2 = \text{left height}$
 $x_3 = \text{right height}$
 $x_4 = \text{bottom frame}$
 $x_5 = \text{top frame}$
 $x_6 = \text{diagonal}.$

Hence, the first PC is essentially the difference between the bottom frame variable and the diagonal. The second PC is best described by the difference between the top frame variable and the sum of bottom frame and diagonal variables.

The weighting of the PCs tells us in which directions, expressed in original coordinates, the best variance explanation is obtained. A measure of how well the first q PCs explain variation is given by the relative proportion:

$$\psi_q = \frac{\sum_{j=1}^q \lambda_j}{\sum_{j=1}^p \lambda_j} = \frac{\sum_{j=1}^q \text{Var}(Y_j)}{\sum_{j=1}^p \text{Var}(Y_j)}. \quad (11.12)$$

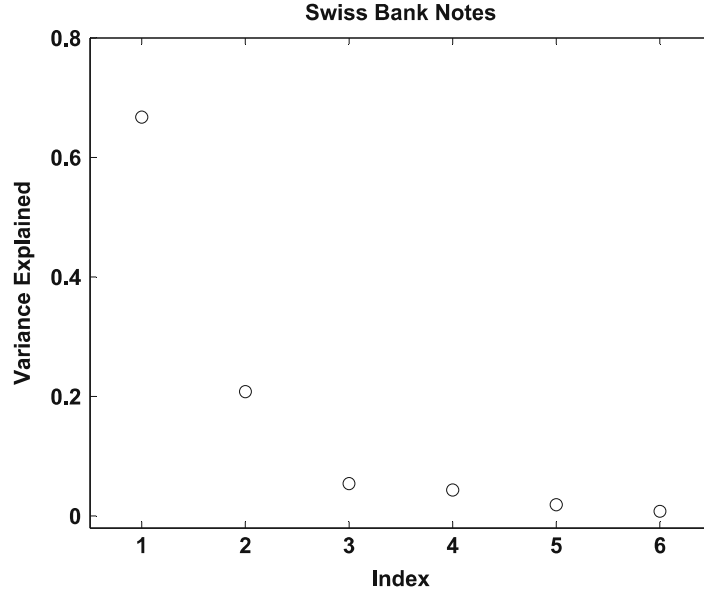
Referring to the bank data Example 11.2, the (cumulative) proportions of explained variance are given in Table 11.1. The first PC ($q = 1$) already explains 67 % of the variation. The first three ($q = 3$) PCs explain 93 % of the variation. Once again it should be noted that PCs are not scale invariant, e.g. the PCs derived from the correlation matrix give different results than the PCs derived from the covariance matrix (see Sect. 11.5).

A good graphical representation of the ability of the PCs to explain the variation in the data is given by the scree plot shown in the lower right-hand window of Fig. 11.3. The screeplot can be modified by using the relative proportions on the y-axis, as is shown in Fig. 11.5 for the bank data set.

Table 11.1 Proportion of variance of PC's

Eigenvalue	Proportion of variance	Cumulated proportion
2.985	0.67	0.67
0.931	0.21	0.88
0.242	0.05	0.93
0.194	0.04	0.97
0.085	0.02	0.99
0.035	0.01	1.00

Fig. 11.5 Relative proportion of variance explained by PCs 
MVApCabanki



The covariance between the PC vector Y and the original vector X is calculated with the help of (11.4) as follows:

$$\begin{aligned}
 \text{Cov}(X, Y) &= E(XY^\top) - E X E Y^\top = E(XY^\top) \\
 &= E(XX^\top \Gamma) - \mu \mu^\top \Gamma = \text{Var}(X) \Gamma \\
 &= \Sigma \Gamma \\
 &= \Gamma \Lambda \Gamma^\top \Gamma \\
 &= \Gamma \Lambda.
 \end{aligned} \tag{11.13}$$

Hence, the correlation, $\rho_{X_i Y_j}$, between variable X_i and the PC Y_j is

$$\rho_{X_i Y_j} = \frac{\gamma_{ij} \lambda_j}{(\sigma_{X_i X_i} \lambda_j)^{1/2}} = \gamma_{ij} \left(\frac{\lambda_j}{\sigma_{X_i X_i}} \right)^{1/2}. \tag{11.14}$$

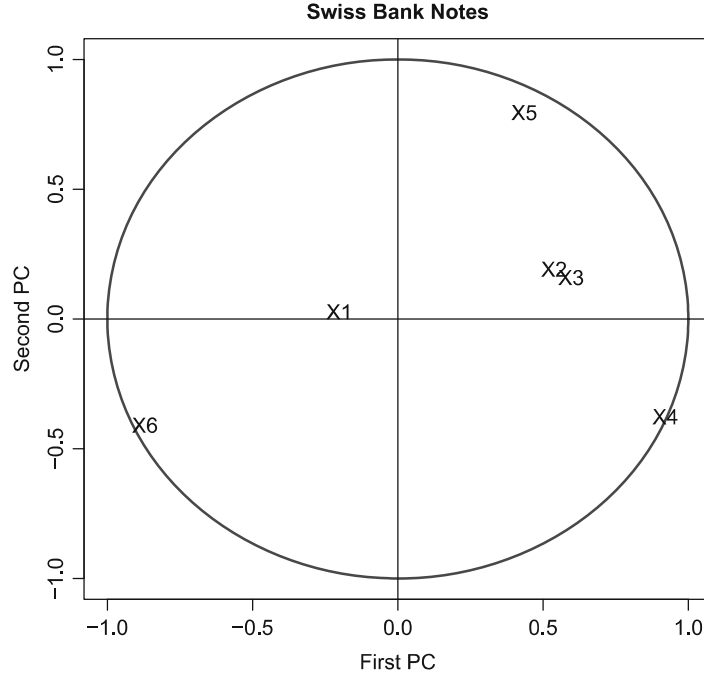
Using actual data, this of course translates into

$$r_{X_i Y_j} = g_{ij} \left(\frac{\ell_j}{s_{X_i X_i}} \right)^{1/2}. \tag{11.15}$$

The correlations can be used to evaluate the relations between the PCs Y_j where $j = 1, \dots, q$, and the original variables X_i where $i = 1, \dots, p$. Note that

$$\sum_{j=1}^p r_{X_i Y_j}^2 = \frac{\sum_{j=1}^p \ell_j g_{ij}^2}{s_{X_i X_i}} = \frac{s_{X_i X_i}}{s_{X_i X_i}} = 1. \tag{11.16}$$

Fig. 11.6 The correlation of the original variable with the PCs  MVApcabank.i



Indeed, $\sum_{j=1}^p \ell_j g_{ij}^2 = g_i^\top \mathcal{L} g_i$ is the (i, i) -element of the matrix $\mathcal{G} \mathcal{L} \mathcal{G}^\top = \mathcal{S}$, so that $r_{X_i Y_j}^2$ may be seen as the proportion of variance of X_i explained by Y_j .

In the space of the first two PCs we plot these proportions, i.e. $r_{X_i Y_1}$ versus $r_{X_i Y_2}$. Figure 11.6 shows this for the bank notes example. This plot shows which of the original variables are most strongly correlated with PC Y_1 and Y_2 .

From (11.16) it obviously follows that $r_{X_i Y_1}^2 + r_{X_i Y_2}^2 \leq 1$ so that the points are always inside the circle of radius 1. In the bank notes example, the variables X_4 , X_5 and X_6 correspond to correlations near the periphery of the circle and are thus well explained by the first two PCs. Recall that we have interpreted the first PC as being essentially the difference between X_4 and X_6 . This is also reflected in Fig. 11.6 since the points corresponding to these variables lie on different sides of the vertical axis. An analogous remark applies to the second PC. We had seen that the second PC is well described by the difference between X_5 and the sum of X_4 and X_6 . Now we are able to see this result again from Fig. 11.6 since the point corresponding to X_5 lies above the horizontal axis and the points corresponding to X_4 and X_6 lie below.

The correlations of the original variables X_i and the first two PCs are given in Table 11.2 along with the cumulated percentage of variance of each variable explained by Y_1 and Y_2 . This table confirms the above results. In particular, it confirms that the percentage of variance of X_1 (and X_2 , X_3) explained by the first two PCs is relatively small and so are their weights in the graphical representation of the individual bank notes in the space of the first two PCs (as can be seen in the upper left plot in Fig. 11.3). Looking simultaneously at Fig. 11.6 and the upper left plot of Fig. 11.3 shows that the genuine bank notes are roughly characterised by large values of X_6 and smaller values of X_4 . The counterfeit bank notes show larger values of X_5 (see Example 7.15).

Table 11.2 Correlation between the original variables and the PCs

	$r_{X_1 Y_1}$	$r_{X_1 Y_2}$	$r_{X_1 Y_1}^2 + r_{X_1 Y_2}^2$
X_1 length	-0.201	0.028	0.041
X_2 left h.	0.538	0.191	0.326
X_3 right h.	0.597	0.159	0.381
X_4 lower	0.921	-0.377	0.991
X_5 upper	0.435	0.794	0.820
X_6 diagonal	-0.870	-0.410	0.926



Summary

- ↪ The weighting of the PCs tells us in which directions, expressed in original coordinates, the best explanation of the variance is obtained. Note that the PCs are not scale invariant.
- ↪ A measure of how well the first q PCs explain variation is given by the relative proportion $\psi_q = \sum_{j=1}^q \lambda_j / \sum_{j=1}^p \lambda_j$. A good graphical representation of the ability of the PCs to explain the variation in the data is the scree plot of these proportions.
- ↪ The correlation between PC Y_j and an original variable X_i is $\rho_{X_i Y_j} = \gamma_{ij} \left(\frac{\lambda_j}{\sigma_{X_i X_i}} \right)^{1/2}$. For a data matrix this translates into $r_{X_i Y_j}^2 = \frac{\ell_j g_{ij}^2}{s_{X_i X_i}} \cdot r_{X_i Y_j}^2$ can be interpreted as the proportion of variance of X_i explained by Y_j . A plot of $r_{X_i Y_1}$ vs. $r_{X_i Y_2}$ shows which of the original variables are most strongly correlated with the PCs, namely those that are close to the periphery of the circle of radius 1.

11.4 Asymptotic Properties of the PCs

In practice, PCs are computed from sample data. The following theorem yields results on the asymptotic distribution of the sample PCs.

Theorem 11.4 Let $\Sigma > 0$ with distinct eigenvalues, and let $\mathcal{U} \sim m^{-1} W_p(\Sigma, m)$ with spectral decompositions $\Sigma = \Gamma \Lambda \Gamma^\top$, and $\mathcal{U} = \mathcal{G} \mathcal{L} \mathcal{G}^\top$. Then

- (a) $\sqrt{m}(\ell - \lambda) \xrightarrow{\mathcal{L}} N_p(0, 2\Lambda^2)$,
where $\ell = (\ell_1, \dots, \ell_p)^\top$ and $\lambda = (\lambda_1, \dots, \lambda_p)^\top$ are the diagonals of $\mathcal{L}$ and Λ ,
- (b) $\sqrt{m}(g_j - \gamma_j) \xrightarrow{\mathcal{L}} N_p(0, \mathcal{V}_j)$,
with $\mathcal{V}_j = \lambda_j \sum_{k \neq j} \frac{\lambda_k}{(\lambda_k - \lambda_j)^2} \gamma_k \gamma_k^\top$,

(c) $\text{Cov}(g_j, g_k) = \mathcal{V}_{jk}$,

where the (r, s) -element of the matrix $\mathcal{V}_{jk}(p \times p)$ is $-\frac{\lambda_j \lambda_k \gamma_{rk} \gamma_{sj}}{m(\lambda_j - \lambda_k)^2}$,

(d) the elements in ℓ are asymptotically independent of the elements in $\mathcal{G}$.

Example 11.4 Since $n\mathcal{S} \sim W_p(\Sigma, n-1)$ if $X_1, \dots, X_n$ are drawn from $N(\mu, \Sigma)$, we have that

$$\sqrt{n-1}(\ell_j - \lambda_j) \xrightarrow{\mathcal{L}} N(0, 2\lambda_j^2), \quad j = 1, \dots, p. \quad (11.17)$$

Since the variance of (11.17) depends on the true mean λ_j a log transformation is useful. Consider $f(\ell_j) = \log(\ell_j)$. Then $\frac{d}{d\ell_j} f|_{\ell_j=\lambda_j} = \frac{1}{\lambda_j}$ and by the Transformation Theorem 4.11 we have from (11.17) that

$$\sqrt{n-1}(\log \ell_j - \log \lambda_j) \xrightarrow{\mathcal{L}} N(0, 2). \quad (11.18)$$

Hence,

$$\sqrt{\frac{n-1}{2}} (\log \ell_j - \log \lambda_j) \xrightarrow{\mathcal{L}} N(0, 1)$$

and a two-sided confidence interval at the $1 - \alpha = 0.95$ significance level is given by

$$\log(\ell_j) - 1.96\sqrt{\frac{2}{n-1}} \leq \log \lambda_j \leq \log(\ell_j) + 1.96\sqrt{\frac{2}{n-1}}.$$

In the bank data example we have that

$$\ell_1 = 2.98.$$

Therefore,

$$\log(2.98) \pm 1.96\sqrt{\frac{2}{199}} = \log(2.98) \pm 0.1965.$$

It can be concluded for the true eigenvalue that

$$P\{\lambda_1 \in (2.448, 3.62)\} \approx 0.95.$$

Variance Explained by the First q PCs

The variance explained by the first q PCs is given by

$$\psi = \frac{\lambda_1 + \cdots + \lambda_q}{\sum_{j=1}^p \lambda_j}.$$

In practice this is estimated by

$$\hat{\psi} = \frac{\ell_1 + \cdots + \ell_q}{\sum_{j=1}^p \ell_j}.$$

From Theorem 11.4 we know the distribution of $\sqrt{n-1}(\ell - \lambda)$. Since ψ is a non-linear function of λ , we can again apply the Transformation Theorem 4.11 to obtain that

$$\sqrt{n-1}(\hat{\psi} - \psi) \xrightarrow{\mathcal{L}} N(0, \mathcal{D}^\top \mathcal{V} \mathcal{D})$$

where $\mathcal{V} = 2\Lambda^2$ (from Theorem 11.4) and $\mathcal{D} = (d_1, \dots, d_p)^\top$ with

$$d_j = \frac{\partial \psi}{\partial \lambda_j} = \begin{cases} \frac{1 - \psi}{\text{tr}(\Sigma)} & \text{for } 1 \leq j \leq q, \\ \frac{-\psi}{\text{tr}(\Sigma)} & \text{for } q+1 \leq j \leq p. \end{cases}$$

Given this result, the following theorem can be derived.

Theorem 11.5

$$\sqrt{n-1}(\hat{\psi} - \psi) \xrightarrow{\mathcal{L}} N(0, \omega^2),$$

where

$$\begin{aligned} \omega^2 &= \mathcal{D}^\top \mathcal{V} \mathcal{D} = \frac{2}{\{\text{tr}(\Sigma)\}^2} \left\{ (1 - \psi)^2 (\lambda_1^2 + \cdots + \lambda_q^2) + \psi^2 (\lambda_{q+1}^2 + \cdots + \lambda_p^2) \right\} \\ &= \frac{2 \text{tr}(\Sigma^2)}{\{\text{tr}(\Sigma)\}^2} (\psi^2 - 2\beta\psi + \beta) \end{aligned}$$

and

$$\beta = \frac{\lambda_1^2 + \cdots + \lambda_q^2}{\lambda_1^2 + \cdots + \lambda_p^2}.$$

Example 11.5 From Sect. 11.3 it is known that the first PC for the Swiss bank notes resolves 67 % of the variation. It can be tested whether the true proportion is actually 75 %. Computing

$$\hat{\beta} = \frac{\ell_1^2}{\ell_1^2 + \dots + \ell_p^2} = \frac{(2.985)^2}{(2.985)^2 + (0.931)^2 + \dots + (0.035)^2} = 0.902$$

$$\text{tr}(\mathcal{S}) = 4.472$$

$$\text{tr}(\mathcal{S}^2) = \sum_{j=1}^p \ell_j^2 = 9.883$$

$$\begin{aligned} \hat{\omega}^2 &= \frac{2 \text{tr}(\mathcal{S}^2)}{\{\text{tr}(\mathcal{S})\}^2} (\hat{\psi}^2 - 2\hat{\beta}\hat{\psi} + \hat{\beta}) \\ &= \frac{2 \cdot 9.883}{(4.472)^2} \{(0.668)^2 - 2(0.902)(0.668) + 0.902\} = 0.142. \end{aligned}$$

Hence, a confidence interval at a significance of level $1 - \alpha = 0.95$ is given by

$$0.668 \pm 1.96 \sqrt{\frac{0.142}{199}} = (0.615, 0.720).$$

Clearly the hypothesis that $\psi = 75\%$ can be rejected!



Summary

- ↪ The eigenvalues ℓ_j and eigenvectors g_j are asymptotically, normally distributed, in particular $\sqrt{n-1}(\ell - \lambda) \xrightarrow{\mathcal{L}} N_p(0, 2\Lambda^2)$.
- ↪ For the eigenvalues it holds that $\sqrt{\frac{n-1}{2}} (\log \ell_j - \log \lambda_j) \xrightarrow{\mathcal{L}} N(0, 1)$.
- ↪ Given an asymptotic, normal distribution approximate confidence intervals and tests can be constructed for the proportion of variance which is explained by the first q PCs. The two-sided confidence interval at the $1 - \alpha = 0.95$ level is given by $\log(\ell_j) - 1.96 \sqrt{\frac{2}{n-1}} \leq \log \lambda_j \leq \log(\ell_j) + 1.96 \sqrt{\frac{2}{n-1}}$.

Summary (continued)	
$\hookrightarrow$	It holds for $\hat{\psi}$, the estimate of ψ (the proportion of the variance explained by the first q PCs) that $\sqrt{n-1}(\hat{\psi} - \psi) \xrightarrow{\mathcal{L}} N(0, \omega^2)$, where ω is given in Theorem 11.5.

11.5 Normalised Principal Components Analysis

In certain situations the original variables can be heterogeneous w.r.t. their variances. This is particularly true when the variables are measured on heterogeneous scales (such as years, kilograms, dollars, ...). In this case a description of the information contained in the data needs to be provided which is robust w.r.t. the choice of scale. This can be achieved through a standardisation of the variables, namely

$$\mathcal{X}_S = \mathcal{H}\mathcal{X}\mathcal{D}^{-1/2} \quad (11.19)$$

where $\mathcal{D} = \text{diag}(s_{X_1 X_1}, \dots, s_{X_p X_p})$. Note that $\bar{x}_S = 0$ and $\mathcal{S}_{\mathcal{X}_S} = \mathcal{R}$, the correlation matrix of $\mathcal{X}$. The PC transformations of the matrix $\mathcal{X}_S$ are referred to as the *Normalised Principal Components* (NPCs). The spectral decomposition of $\mathcal{R}$ is

$$\mathcal{R} = \mathcal{G}_{\mathcal{R}} \mathcal{L}_{\mathcal{R}} \mathcal{G}_{\mathcal{R}}^{\top}, \quad (11.20)$$

where $\mathcal{L}_{\mathcal{R}} = \text{diag}(\ell_1^{\mathcal{R}}, \dots, \ell_p^{\mathcal{R}})$ and $\ell_1^{\mathcal{R}} \geq \dots \geq \ell_p^{\mathcal{R}}$ are the eigenvalues of $\mathcal{R}$ with corresponding eigenvectors $g_1^{\mathcal{R}}, \dots, g_p^{\mathcal{R}}$ (note that here $\sum_{j=1}^p \ell_j^{\mathcal{R}} = \text{tr}(\mathcal{R}) = p$).

The NPCs, Z_j , provide a representation of each individual, and is given by

$$\mathcal{Z} = \mathcal{X}_S \mathcal{G}_{\mathcal{R}} = (z_1, \dots, z_p). \quad (11.21)$$

After transforming the variables, once again, we have that

$$\bar{z} = 0, \quad (11.22)$$

$$\mathcal{S}_{\mathcal{Z}} = \mathcal{G}_{\mathcal{R}}^{\top} \mathcal{S}_{\mathcal{X}_S} \mathcal{G}_{\mathcal{R}} = \mathcal{G}_{\mathcal{R}}^{\top} \mathcal{R} \mathcal{G}_{\mathcal{R}} = \mathcal{L}_{\mathcal{R}}. \quad (11.23)$$



The NPCs provide a perspective similar to that of the PCs, but in terms of the relative position of individuals, NPC gives each variable the same weight (with the PCs the variable with the largest variance received the largest weight).

Computing the covariance and correlation between X_i and Z_j is straightforward:

$$\mathcal{S}_{X_S, Z} = \frac{1}{n} \mathcal{X}_S^\top \mathcal{Z} = \mathcal{G}_R \mathcal{L}_R, \quad (11.24)$$

$$\mathcal{R}_{X_S, Z} = \mathcal{G}_R \mathcal{L}_R \mathcal{L}_R^{-1/2} = \mathcal{G}_R \mathcal{L}_R^{1/2}. \quad (11.25)$$

The correlations between the original variables X_i and the NPCs Z_j are:

$$r_{X_i Z_j} = \sqrt{\ell_j} g_{R, ij} \quad (11.26)$$

$$\sum_{j=1}^p r_{X_i Z_j}^2 = 1 \quad (11.27)$$

(compare this to (11.15) and (11.16)). The resulting NPCs, the Z_j , can be interpreted in terms of the original variables and the role of each PC in explaining the variation in variable X_i can be evaluated.

11.6 Principal Components as a Factorial Method

The empirical PCs (normalised or not) turn out to be equivalent to the factors that one would obtain by decomposing the appropriate data matrix into its factors (see Chap. 10). It will be shown that the PCs are the factors representing the rows of the centred data matrix and that the NPCs correspond to the factors of the standardised data matrix. The representation of the columns of the standardised data matrix provides (at a scale factor) the correlations between the NPCs and the original variables. The derivation of the (N)PCs presented above will have a nice geometric justification here since they are the best fit in subspaces generated by the columns of the (transformed) data matrix $\mathcal{X}$. This analogy provides complementary interpretations of the graphical representations shown above.

Assume, as in Chap. 10, that we want to obtain representations of the individuals (the rows of $\mathcal{X}$) and of the variables (the columns of $\mathcal{X}$) in spaces of smaller dimension. To keep the representations simple, some prior transformations are performed. Since the origin has no particular statistical meaning in the space of individuals, we will first shift the origin to the centre of gravity, $\bar{x}$, of the point cloud. This is the same as analysing the centred data matrix $\mathcal{X}_C = \mathcal{H}\mathcal{X}$. Now all of the variables have zero means, thus the technique used in Chap. 10 can be applied to the matrix $\mathcal{X}_C$. Note that the spectral decomposition of $\mathcal{X}_C^\top \mathcal{X}_C$ is related to that of $\mathcal{S}_X$, namely

$$\mathcal{X}_C^\top \mathcal{X}_C = \mathcal{X}^\top \mathcal{H}^\top \mathcal{H} \mathcal{X} = n\mathcal{S}_X = n\mathcal{G}\mathcal{L}\mathcal{G}^\top. \quad (11.28)$$

The factorial variables are obtained by projecting $\mathcal{X}_C$ on $\mathcal{G}$,

$$\mathcal{Y} = \mathcal{X}_C \mathcal{G} = (y_1, \dots, y_p). \quad (11.29)$$

These are the same principal components obtained above, see formula (11.10). (Note that the y 's here correspond to the z 's in Sect. 10.2.) Since $\mathcal{H}\mathcal{X}_C = \mathcal{X}_C$, it immediately follows that

$$\bar{y} = 0, \quad (11.30)$$

$$\mathcal{S}_Y = \mathcal{G}^\top \mathcal{S}_X \mathcal{G} = \mathcal{L} = \text{diag}(\ell_1, \dots, \ell_p). \quad (11.31)$$

The scatterplot of the individuals on the factorial axes are thus centred around the origin and are more spread out in the first direction (first PC has variance ℓ_1) than in the second direction (second PC has variance ℓ_2).

The representation of the variables can be obtained using the Duality Relations (10.11), and (10.12). The projections of the columns of $\mathcal{X}_C$ onto the eigenvectors v_k of $\mathcal{X}_C \mathcal{X}_C^\top$ are

$$\mathcal{X}_C^\top v_k = \frac{1}{\sqrt{n\ell_k}} \mathcal{X}_C^\top \mathcal{X}_C g_k = \sqrt{n\ell_k} g_k. \quad (11.32)$$

Thus the projections of the variables on the first p axes are the columns of the matrix

$$\mathcal{X}_C^\top \mathcal{V} = \sqrt{n} \mathcal{G} \mathcal{L}^{1/2}. \quad (11.33)$$

Considering the geometric representation, there is a nice statistical interpretation of the angle between two columns of $\mathcal{X}_C$. Given that

$$x_{C[j]}^\top x_{C[k]} = ns_{X_j X_k}, \quad (11.34)$$

$$\|x_{C[j]}\|^2 = ns_{X_j X_j}, \quad (11.35)$$

where $x_{C[j]}$ and $x_{C[k]}$ denote the j -th and k -th column of $\mathcal{X}_C$, it holds that in the full space of the variables, if θ_{jk} is the angle between two variables, $x_{C[j]}$ and $x_{C[k]}$, then

$$\cos \theta_{jk} = \frac{x_{C[j]}^\top x_{C[k]}}{\|x_{C[j]}\| \|x_{C[k]}\|} = r_{X_j X_k}. \quad (11.36)$$

(Example 2.11 shows the general connection that exists between the angle and correlation of two variables). As a result, the relative positions of the variables in the scatterplot of the first columns of $\mathcal{X}_C^\top \mathcal{V}$ may be interpreted in terms of their correlations; the plot provides a picture of the correlation structure of the original data set. Clearly, one should take into account the percentage of variance explained by the chosen axes when evaluating the correlation.

The NPCs can also be viewed as a factorial method for reducing the dimension. The variables are again standardised so that each one has mean zero and unit variance and is independent of the scale of the variables. The factorial analysis of $\mathcal{X}_S$ provides the NPCs. The spectral decomposition of $\mathcal{X}_S^\top \mathcal{X}_S$ is related to that of $\mathcal{R}$, namely

$$\mathcal{X}_S^\top \mathcal{X}_S = \mathcal{D}^{-1/2} \mathcal{X}^\top \mathcal{H} \mathcal{X} \mathcal{D}^{-1/2} = n\mathcal{R} = n\mathcal{G}_{\mathcal{R}} \mathcal{L}_{\mathcal{R}} \mathcal{G}_{\mathcal{R}}^\top.$$

The NPCs Z_j , given by (11.21), may be viewed as the projections of the rows of $\mathcal{X}_S$ onto $\mathcal{G}_{\mathcal{R}}$.

The representation of the variables are again given by the columns of

$$\mathcal{X}_S^\top \mathcal{V}_{\mathcal{R}} = \sqrt{n} \mathcal{G}_{\mathcal{R}} \mathcal{L}_{\mathcal{R}}^{1/2}. \quad (11.37)$$

Comparing (11.37) and (11.25) we see that the projections of the variables in the factorial analysis provide the correlation between the NPCs Z_k and the original variables $x_{[j]}$ (up to the factor $\sqrt{n}$ which could be the scale of the axes).

This implies that a deeper interpretation of the representation of the individuals can be obtained by looking simultaneously at the graphs plotting the variables. Note that

$$x_{S[j]}^\top x_{S[k]} = n r_{X_j X_k}, \quad (11.38)$$

$$\|x_{S[j]}\|^2 = n, \quad (11.39)$$

where $x_{S[j]}$ and $x_{S[k]}$ denote the j -th and k -th column of $\mathcal{X}_S$. Hence, in the full space, all the standardised variables (columns of $\mathcal{X}_S$) are contained within the “sphere” in $\mathbb{R}^n$, which is centred at the origin and has radius $\sqrt{n}$ (the scale of the graph). As in (11.36), given the angle θ_{jk} between two columns $x_{S[j]}$ and $x_{S[k]}$, it holds that

$$\cos \theta_{jk} = r_{X_j X_k}. \quad (11.40)$$

Therefore, when looking at the representation of the variables in the spaces of reduced dimension (for instance the first two factors), we have a picture of the correlation structure between the original X_i ’s in terms of their angles. Of course, the quality of the representation in those subspaces has to be taken into account, which is presented in the next section.

Quality of the Representations

As said before, an overall measure of the quality of the representation is given by

$$\psi = \frac{\ell_1 + \ell_2 + \cdots + \ell_q}{\sum_{j=1}^p \ell_j}.$$

In practice, q is chosen to be equal to 1, 2 or 3. Suppose for instance that $\psi = 0.93$ for $q = 2$. This means that the graphical representation in two dimensions captures 93 % of the total variance. In other words, there is minimal dispersion in a third direction (no more than 7 %).

It can be useful to check if each individual is well represented by the PCs. Clearly, the proximity of two individuals on the projected space may not necessarily coincide with the proximity in the full original space $\mathbb{R}^p$, which may lead to erroneous interpretations of the graphs. In this respect, it is worth computing the angle ϑ_{ik} between the representation of an individual i and the k -th PC or NPC axis. This can be done using (2.40), i.e.

$$\cos \vartheta_{ik} = \frac{y_i^\top e_k}{\|y_i\| \|e_k\|} = \frac{y_{ik}}{\|x_{Ci}\|}$$

for the PCs or analogously

$$\cos \zeta_{ik} = \frac{z_i^\top e_k}{\|z_i\| \|e_k\|} = \frac{z_{ik}}{\|x_{Si}\|}$$

for the NPCs, where e_k denotes the k -th unit vector $e_k = (0, \dots, 1, \dots, 0)^\top$. An individual i will be represented on the k -th PC axis if its corresponding angle is small, i.e. if $\cos^2 \vartheta_{ik}$ for $k = 1, \dots, p$ is close to one. Note that for each individual i ,

$$\sum_{k=1}^p \cos^2 \vartheta_{ik} = \frac{y_i^\top y_i}{x_{Ci}^\top x_{Ci}} = \frac{x_{Ci}^\top \mathcal{G} \mathcal{G}^\top x_{Ci}}{x_{Ci}^\top x_{Ci}} = 1.$$

The values $\cos^2 \vartheta_{ik}$ are sometimes called the relative contributions of the k -th axis to the representation of the i -th individual, e.g. if $\cos^2 \vartheta_{i1} + \cos^2 \vartheta_{i2}$ is large (near one), we know that the individual i is well represented on the plane of the first two principal axes since its corresponding angle with the plane is close to zero.

We already know that the quality of the representation of the variables can be evaluated by the percentage of X_i 's variance that is explained by a PC, which is given by $r_{X_i Y_j}^2$ or $r_{X_i Z_j}^2$ according to (11.16) and (11.27) respectively.

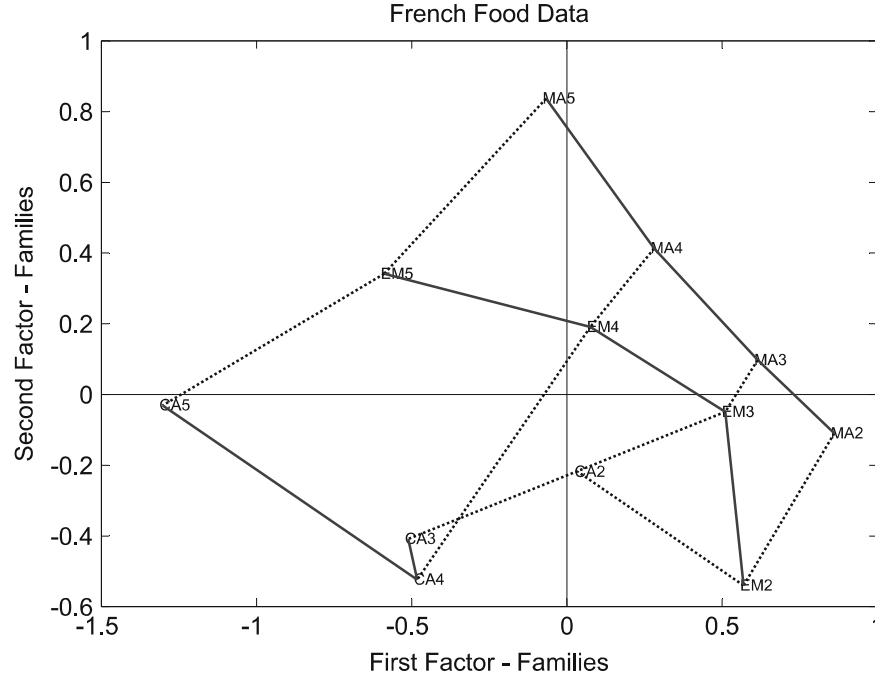


Fig. 11.7 Representation of the individuals  MVAnpcafood

Example 11.6 Let us return to the French food expenditure example, see Sect. 22.6. This yields a two-dimensional representation of the individuals as shown in Fig. 11.7.

Calculating the matrix $\mathcal{G}_{\mathcal{R}}$ we have

$$\mathcal{G}_{\mathcal{R}} = \begin{pmatrix} -0.240 & 0.622 & -0.011 & -0.544 & 0.036 & 0.508 \\ -0.466 & 0.098 & -0.062 & -0.023 & -0.809 & -0.301 \\ -0.446 & -0.205 & 0.145 & 0.548 & -0.067 & 0.625 \\ -0.462 & -0.141 & 0.207 & -0.053 & 0.411 & -0.093 \\ -0.438 & -0.197 & 0.356 & -0.324 & 0.224 & -0.350 \\ -0.281 & 0.523 & -0.444 & 0.450 & 0.341 & -0.332 \\ 0.206 & 0.479 & 0.780 & 0.306 & -0.069 & -0.138 \end{pmatrix},$$

which gives the weights of the variables (milk, vegetables, etc.). The eigenvalues ℓ_j and the proportions of explained variance are given in Table 11.3.

The interpretation of the principal components are best understood when looking at the correlations between the original X_i 's and the PCs. Since the first two PCs explain 88.1 % of the variance, we limit ourselves to the first two PCs. The results are shown in Table 11.4. The two-dimensional graphical representation of the variables in Fig. 11.8 is based on the first two columns of Table 11.4.

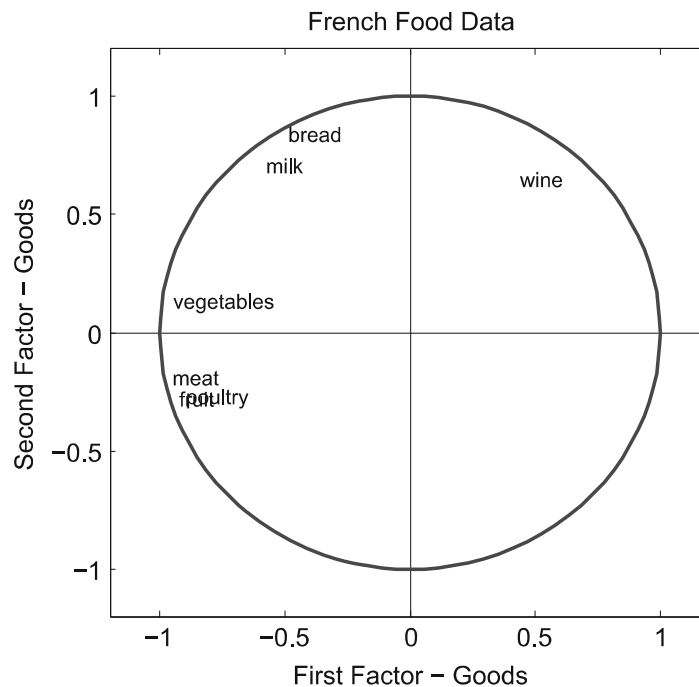
The plots are the projections of the variables into $\mathbb{R}^2$. Since the quality of the representation is good for all the variables (except maybe X_7), their relative angles give a picture of their original correlation: wine is negatively correlated with the

Table 11.3 Eigenvalues and explained variance

Eigenvalues	Proportion of variance	Cumulated proportion
4.333	0.6190	61.9
1.830	0.2620	88.1
0.631	0.0900	97.1
0.128	0.0180	98.9
0.058	0.0080	99.7
0.019	0.0030	99.9
0.001	0.0001	100.0

Table 11.4 Correlations with PCs

	$r_{X_i Z_1}$	$r_{X_i Z_2}$	$r_{X_i Z_1}^2 + r_{X_i Z_2}^2$
X_1 : bread	−0.499	0.842	0.957
X_2 : vegetables	−0.970	0.133	0.958
X_3 : fruits	−0.929	−0.278	0.941
X_4 : meat	−0.962	−0.191	0.962
X_5 : poultry	−0.911	−0.266	0.901
X_6 : milk	−0.584	0.707	0.841
X_7 : wine	0.428	0.648	0.604

Fig. 11.8 Representation of the variables  MVAnpcafood

vegetables, fruits, meat and poultry groups ($\theta > 90^\circ$), whereas taken individually this latter grouping of variables are highly positively correlated with each other ($\theta \approx 0$). Bread and milk are positively correlated but poorly correlated with meat, fruits and poultry ($\theta \approx 90^\circ$).

Now the representation of the individuals in Fig. 11.7 can be interpreted better. From Fig. 11.8 and Table 11.4 we can see that the first factor Z_1 is a vegetable–meat–poultry–fruit factor (with a negative sign), whereas the second

factor is a milk–bread–wine factor (with a positive sign). Note that this corresponds to the most important weights in the first columns of $\mathcal{G}_{\mathcal{R}}$. In Fig. 11.7 lines were drawn to connect families of the same size and families of the same professional types. A grid can clearly be seen (with a slight deformation by the manager families) that shows the families with higher expenditures (higher number of children) on the left.

Considering both figures together explains what types of expenditures are responsible for similarities in food expenditures. Bread, milk and wine expenditures are similar for manual workers and employees. Families of managers are characterised by higher expenditures on vegetables, fruits, meat and poultry. Very often when analysing NPCs (and PCs), it is illuminating to use such a device to introduce qualitative aspects of individuals in order to enrich the interpretations of the graphs.

	Summary
↔	NPCs are PCs applied to the standardised (normalised) data matrix $\mathcal{X}_S$.
↔	The graphical representation of NPCs provides a similar type of picture as that of PCs, the difference being in the relative position of individuals, i.e. each variable in NPCs has the same weight (in PCs, the variable with the largest variance has the largest weight).
↔	The quality of the representation is evaluated by $\psi = (\sum_{j=1}^p \ell_j)^{-1}(\ell_1 + \ell_2 + \dots + \ell_q)$.
↔	The quality of the representation of a variable can be evaluated by the percentage of X_i 's variance that is explained by a PC, i.e. $r_{X_i Y_j}^2$.

11.7 Common Principal Components

In many applications a statistical analysis is simultaneously done for groups of data. In this section a technique is presented that allows us to analyse group elements that have common PCs. From a statistical point of view, estimating PCs simultaneously in different groups will result in a joint dimension reducing transformation. This multi-group PCA, the so-called common principle components analysis (CPCA), yields the joint eigenstructure across groups.

In addition to traditional PCA, the basic assumption of CPCA is that the space spanned by the eigenvectors is identical *across* several groups, whereas variances associated with the components are allowed to vary.

More formally, the hypothesis of common principle components can be stated in the following way (Flury, 1988):

$$H_{\text{CPC}} : \Sigma_i = \Gamma \Lambda_i \Gamma^\top, \quad i = 1, \dots, k$$

where Σ_i is a positive definite $p \times p$ population covariance matrix for every i , $\Gamma = (\gamma_1, \dots, \gamma_p)$ is an orthogonal $p \times p$ transformation matrix and $\Lambda_i = \text{diag}(\lambda_{i1}, \dots, \lambda_{ip})$ is the matrix of eigenvalues. Moreover, assume that all λ_i are distinct.

Let S be the (unbiased) sample covariance matrix of an underlying p -variate normal distribution $N_p(\mu, \Sigma)$ with sample size n . Then the distribution of nS has $n - 1$ degrees of freedom and is known as the Wishart distribution (Muirhead, 1982, p. 86):

$$nS \sim \mathcal{W}_p(\Sigma, n - 1).$$

The density is given in (5.16). Hence, for a given Wishart matrix S_i with sample size n_i , the likelihood function can be written as

$$L(\Sigma_1, \dots, \Sigma_k) = C \prod_{i=1}^k \exp \left[\text{tr} \left\{ -\frac{1}{2} (n_i - 1) \Sigma_i^{-1} S_i \right\} \right] |\Sigma_i|^{-\frac{1}{2}(n_i - 1)} \quad (11.41)$$

where C is a constant independent of the parameters Σ_i . Maximising the likelihood is equivalent to minimising the function

$$g(\Sigma_1, \dots, \Sigma_k) = \sum_{i=1}^k (n_i - 1) \left\{ \log |\Sigma_i| + \text{tr}(\Sigma_i^{-1} S_i) \right\}.$$

Assuming that H_{CPC} holds, i.e. in replacing Σ_i by $\Gamma \Lambda_i \Gamma^\top$, after some manipulations one obtains

$$g(\Gamma, \Lambda_1, \dots, \Lambda_k) = \sum_{i=1}^k (n_i - 1) \sum_{j=1}^p \left(\log \lambda_{ij} + \frac{\gamma_j^\top S_i \gamma_j}{\lambda_{ij}} \right).$$

As we know from Sect. 2.2, the vectors γ_j in Γ have to be orthogonal. Orthogonality of the vectors γ_j is achieved using the Lagrange method, i.e. we impose the p constraints $\gamma_j^\top \gamma_j = 1$ using the Lagrange multipliers μ_j , and the remaining $p(p - 1)/2$ constraints $\gamma_h^\top \gamma_j = 0$ for $h \neq j$ using the multiplier $2\mu_{hj}$ (Flury, 1988). This yields

$$g^*(\Gamma, \Lambda_1, \dots, \Lambda_k) = g(\cdot) - \sum_{j=1}^p \mu_j (\gamma_j^\top \gamma_j - 1) - 2 \sum_{h=1}^p \sum_{j=h+1}^p \mu_{hj} \gamma_h^\top \gamma_j.$$

Taking partial derivatives with respect to all λ_{im} and γ_m , it can be shown that the solution of the CPC model is given by the generalised system of characteristic equations

$$\gamma_m^\top \left\{ \sum_{i=1}^k (n_i - 1) \frac{\lambda_{im} - \lambda_{ij}}{\lambda_{im} \lambda_{ij}} \mathcal{S}_i \right\} \gamma_j = 0, \quad m, j = 1, \dots, p, \quad m \neq j. \quad (11.42)$$

This system can be solved using

$$\lambda_{im} = \gamma_m^\top \mathcal{S}_i \gamma_m, \quad i = 1, \dots, k, \quad m = 1, \dots, p$$

under the constraints

$$\gamma_m^\top \gamma_j = \begin{cases} 0 & m \neq j \\ 1 & m = j \end{cases}.$$

Flury (1988) proves existence and uniqueness of the maximum of the likelihood function, and Flury and Gautschi (1986) provide a numerical algorithm.

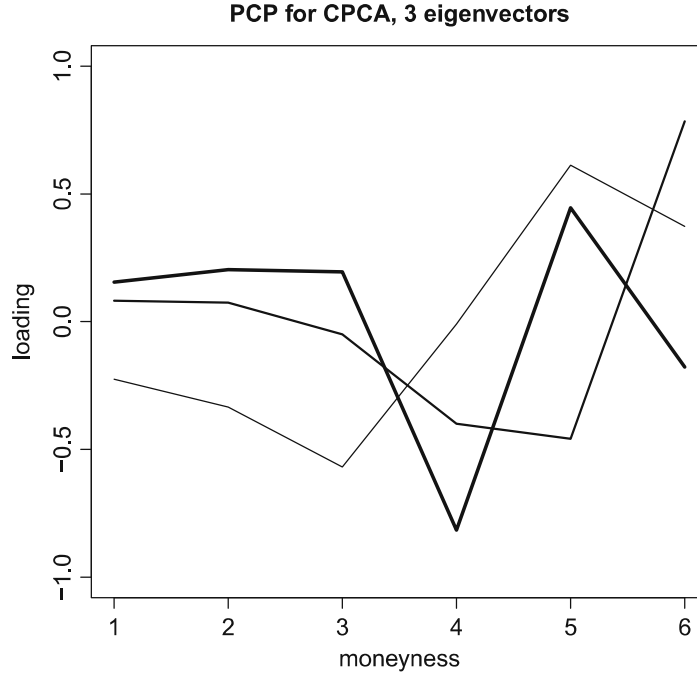
Example 11.7 As an example we provide the data sets XFGvolsurf01, XFGvolsurf02 and XFGvolsurf03 that have been used in Fengler, Härdle, and Villa (2003) to estimate common principle components for the implied volatility surfaces of the DAX 1999. The data has been generated by smoothing an implied volatility surface day by day. Next, the estimated grid points have been grouped into maturities of $\tau = 1$, $\tau = 2$ and $\tau = 3$ months and transformed into a vector of time series of the “smile”, i.e. each element of the vector belongs to a distinct moneyness ranging from 0.85 to 1.10.

Figure 11.9 shows the first three eigenvectors in a parallel coordinate plot. The basic structure of the first three eigenvectors is not altered. We find a shift, a slope and a twist structure. This structure is *common* to all maturity groups, i.e. when exploiting PCA as a dimension reducing tool, the same transformation applies to each group! However, by comparing the size of eigenvalues among groups we find that variability is decreasing across groups as we move from the short-term contracts to long-term contracts.

Before drawing conclusions we should convince ourselves that the CPC model is truly a good description of the data. This can be done by using a likelihood ratio test. The likelihood ratio statistic for comparing a restricted (the CPC) model against the unrestricted model (the model where all covariances are treated separately) is given by

$$T_{(n_1, n_2, \dots, n_k)} = -2 \log \frac{L(\hat{\Sigma}_1, \dots, \hat{\Sigma}_k)}{L(\mathcal{S}_1, \dots, \mathcal{S}_k)}.$$

Fig. 11.9 Factor loadings of the first (*thick*), the second (*medium*), and the third (*thin*) PC  MVAcpcaiv



Inserting the likelihood function, we find that this is equivalent to

$$T_{(n_1, n_2, \dots, n_k)} = \sum_{i=1}^k (n_i - 1) \frac{\det(\hat{\Sigma}_i)}{\det(\mathcal{S}_i)},$$

which has a χ^2 distribution as $\min(n_i)$ tends to infinity with

$$k \left\{ \frac{1}{2} p(p-1) + 1 \right\} - \left\{ \frac{1}{2} p(p-1) + kp \right\} = \frac{1}{2} (k-1) p(p-1)$$

degrees of freedom. This test is included in the quantlet  MVAcpcaiv.

The calculations yield $T_{(n_1, n_2, \dots, n_k)} = 31.836$, which corresponds to the p -value $p = 0.37512$ for the $\chi^2(30)$ distribution. Hence we cannot reject the CPC model against the unrestricted model, where PCA is applied to each maturity separately.

Using the methods in Sect. 11.3, we can estimate the amount of variability, ζ_l , explained by the first l principal components: (only a few factors, three at the most, are needed to capture a large amount of the total variability present in the data). Since the model now captures the variability in both the strike and maturity dimensions, this is a suitable starting point for a simplified VaR calculation for delta-gamma neutral option portfolios using Monte Carlo methods, and is hence a valuable insight in risk management.

11.8 Boston Housing

A set of transformations were defined in Chap. 1 for the Boston Housing data set that resulted in “regular” marginal distributions. The usefulness of principal component analysis with respect to such high-dimensional data sets will now be shown. The variable X_4 is dropped because it is a discrete 0–1 variable. It will be used later, however, in the graphical representations. The scale difference of the remaining 13 variables motivates a NPCA based on the correlation matrix.

The eigenvalues and the percentage of explained variance are given in Table 11.5.

The first principal component explains 56 % of the total variance and the first three components together explain more than 75 %. These results imply that it is sufficient to look at 2, maximum 3, principal components.

Table 11.6 provides the correlations between the first three PCs and the original variables. These can be seen in Fig. 11.10.

The correlations with the first PC show a very clear pattern. The variables X_2 , X_6 , X_8 , X_{12} , and X_{14} are strongly positively correlated with the first PC, whereas the remaining variables are highly negatively correlated. The minimal correlation in the absolute value is 0.5. The first PC axis could be interpreted as a quality of life and house indicator. The second axis, given the polarities of X_{11} and X_{13} and of X_6 and X_{14} , can be interpreted as a social factor explaining only 10 % of the total variance. The third axis is dominated by a polarity between X_2 and X_{12} .

The set of individuals from the first two PCs can be graphically interpreted if the plots are colour coded with respect to some particular variable of interest.

Table 11.5 Eigenvalues and percentage of explained variance for Boston Housing data  MVAncapahousi

Eigenvalue	Percentages	Cumulated percentages
7.2852	0.5604	0.5604
1.3517	0.1040	0.6644
1.1266	0.0867	0.7510
0.7802	0.0600	0.8111
0.6359	0.0489	0.8600
0.5290	0.0407	0.9007
0.3397	0.0261	0.9268
0.2628	0.0202	0.9470
0.1936	0.0149	0.9619
0.1547	0.0119	0.9738
0.1405	0.0108	0.9846
0.1100	0.0085	0.9931
0.0900	0.0069	1.0000

Table 11.6 Correlations of the first three PC's with the original variables MVAncpcahous

	PC ₁	PC ₂	PC ₃
X_1	-0.9076	0.2247	0.1457
X_2	0.6399	-0.0292	0.5058
X_3	-0.8580	0.0409	-0.1845
X_5	-0.8737	0.2391	-0.1780
X_6	0.5104	0.7037	0.0869
X_7	-0.7999	0.1556	-0.2949
X_8	0.8259	-0.2904	0.2982
X_9	-0.7531	0.2857	0.3804
X_{10}	-0.8114	0.1645	0.3672
X_{11}	-0.5674	-0.2667	0.1498
X_{12}	0.4906	-0.1041	-0.5170
X_{13}	-0.7996	-0.4253	-0.0251
X_{14}	0.7366	0.5160	-0.1747

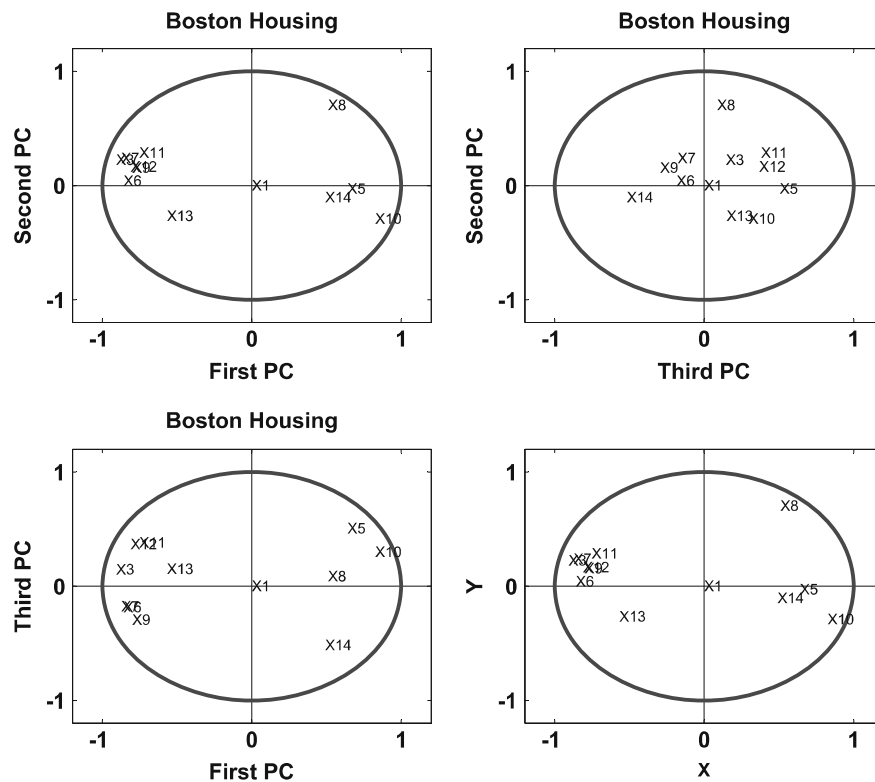


Fig. 11.10 NPCA for the Boston housing data, correlations of first three PCs with the original variables MVAncpcahous

Figure 11.11 colour codes $X_{14} > \text{median}$ as red points. Clearly the first and second PCs are related to house value. The situation is less clear in Fig. 11.12 where the colour code corresponds to X_4 , the Charles River indicator, i.e. houses near the river are coloured red.

Fig. 11.11 NPC analysis for the Boston housing data, scatterplot of the first two PCs. More expensive houses are marked with *red colour*  MVAnpcahous

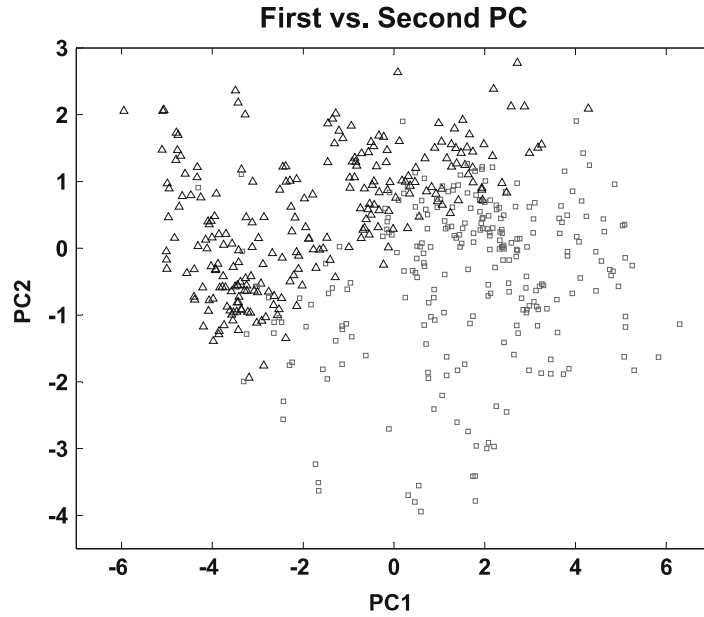
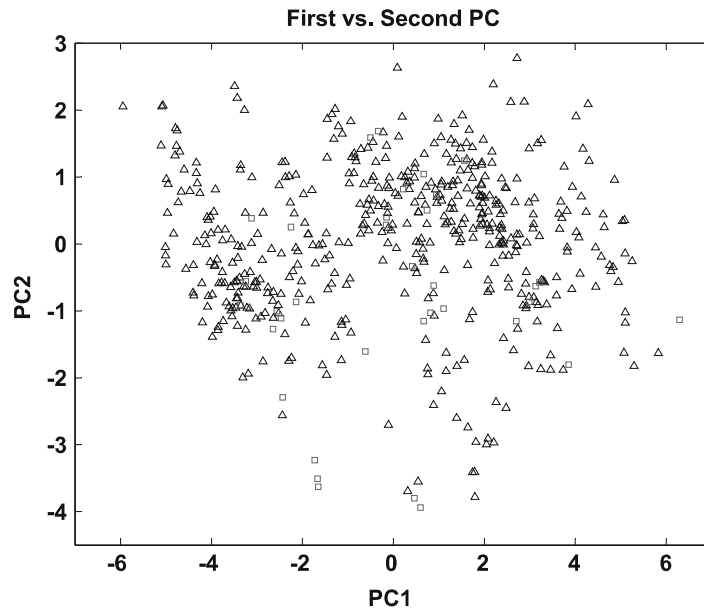


Fig. 11.12 NPC analysis for the Boston housing data, scatterplot of the first two PCs. Houses close to the Charles River are indicated with *red squares*  MVAnpcahous



11.9 More Examples

Example 11.8 Let us now apply the PCA to the *standardised* bank data set (Sect. 22.2). Figure 11.13 shows some PC plots of the bank data set. The genuine and counterfeit bank notes are marked by “o” and “+”, respectively.

The vector of eigenvalues of $\mathcal{R}$ is

$$\ell = (2.946, 1.278, 0.869, 0.450, 0.269, 0.189)^\top.$$

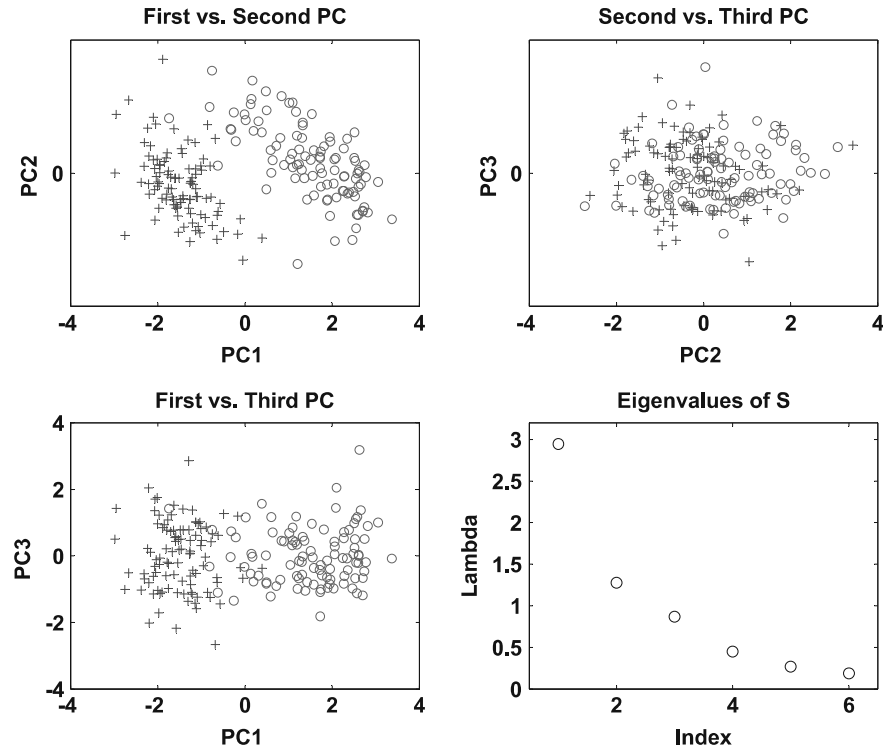


Fig. 11.13 Principal components of the *standardised* bank data  `MVAnpcabank`

Table 11.7 Eigenvalues and proportions of explained variance

ℓ_j	Proportion of variances	Cumulated proportion
2.946	0.491	49.1
1.278	0.213	70.4
0.869	0.145	84.9
0.450	0.075	92.4
0.264	0.045	96.9
0.189	0.032	100.0

The eigenvectors g_j are given by the columns of the matrix

$$\mathcal{G} = \begin{pmatrix} -0.007 & -0.815 & 0.018 & 0.575 & 0.059 & 0.031 \\ 0.468 & -0.342 & -0.103 & -0.395 & -0.639 & -0.298 \\ 0.487 & -0.252 & -0.123 & -0.430 & 0.614 & 0.349 \\ 0.407 & 0.266 & -0.584 & 0.404 & 0.215 & -0.462 \\ 0.368 & 0.091 & 0.788 & 0.110 & 0.220 & -0.419 \\ -0.493 & -0.274 & -0.114 & -0.392 & 0.340 & -0.632 \end{pmatrix}.$$

Each original variable has the same weight in the analysis and the results are independent of the scale of each variable.

The proportions of explained variance are given in Table 11.7. It can be concluded that the representation in two dimensions should be sufficient. The

Fig. 11.14 The correlations of the original variable with the PCs  MVAncabanki

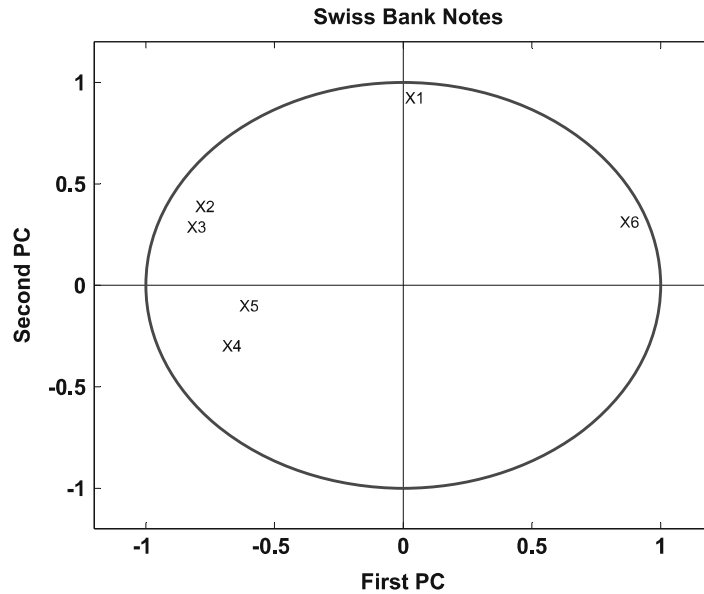


Table 11.8 Correlations with PCs

	$r_{X_i Z_1}$	$r_{X_i Z_2}$	$r_{X_i Z_1}^2 + r_{X_i Z_2}^2$
X_1 : length	−0.012	−0.922	0.85
X_2 : left height	0.803	−0.387	0.79
X_3 : right height	0.835	−0.285	0.78
X_4 : lower	0.698	0.301	0.58
X_5 : upper	0.631	0.104	0.41
X_6 : diagonal	−0.847	−0.310	0.81

correlations leading to Fig. 11.14 are given in Table 11.8. The picture is different from the one obtained in Sect. 11.3 (see Table 11.2). Here, the first factor is mainly a left–right vs. diagonal factor and the second one is a length factor (with negative weight). Take another look at Fig. 11.13, where the individual bank notes are displayed. In the upper left graph it can be seen that the genuine bank notes are for the most part in the south-eastern portion of the graph featuring a larger diagonal, smaller height ($Z_1 < 0$) and also a larger length ($Z_2 < 0$). Note also that Fig. 11.14 gives an idea of the correlation structure of the original data matrix.

Example 11.9 Consider the data of 79 US companies given in Table 22.5. The data is first standardised by subtracting the mean and dividing by the standard deviation. Note that the data set contains six variables: assets (X_1), sales (X_2), market value (X_3), profits (X_4), cash flow (X_5), number of employees (X_6).

Calculating the corresponding vector of eigenvalues gives

$$\ell = (5.039, 0.517, 0.359, 0.050, 0.029, 0.007)^\top$$

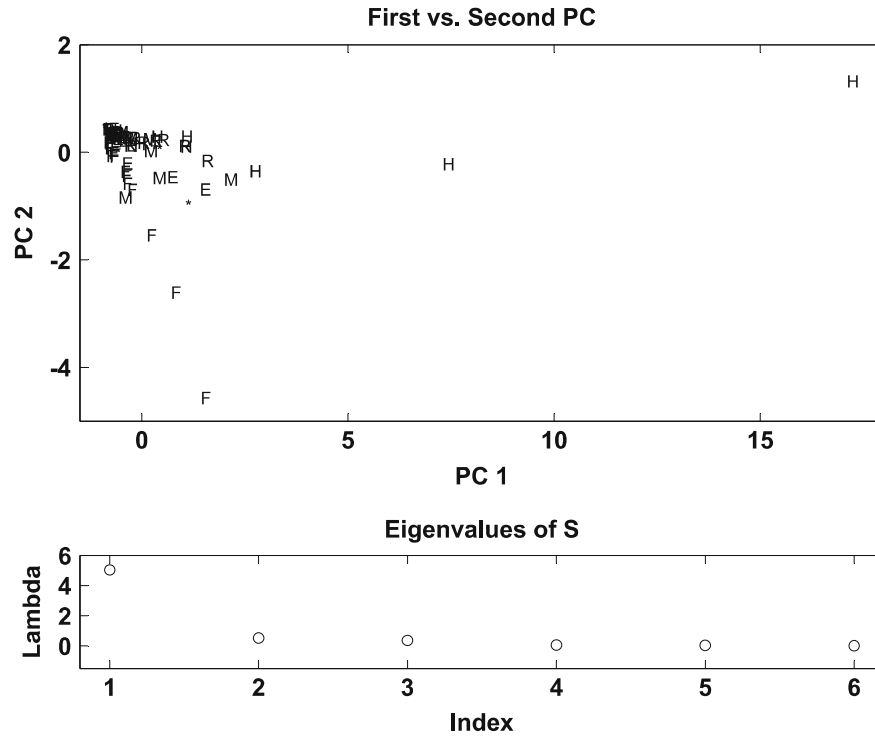


Fig. 11.15 Principal components of the US company data  MVAncpcausco

and the matrix of eigenvectors is

$$\mathcal{G} = \begin{pmatrix} 0.340 & -0.849 & -0.339 & 0.205 & 0.077 & -0.006 \\ 0.423 & -0.170 & 0.379 & -0.783 & -0.006 & -0.186 \\ 0.434 & 0.190 & -0.192 & 0.071 & -0.844 & 0.149 \\ 0.420 & 0.364 & -0.324 & 0.156 & 0.261 & -0.703 \\ 0.428 & 0.285 & -0.267 & -0.121 & 0.452 & 0.667 \\ 0.397 & 0.010 & 0.726 & 0.548 & 0.098 & 0.065 \end{pmatrix}.$$

Using this information the graphical representations of the first two principal components are given in Fig.11.15. The different sectors are marked by the following symbols:

- H ... Hi Tech and Communication
- E ... Energy
- F ... Finance
- M ... Manufacturing
- R ... Retail
- ★ ... all other sectors.

The two outliers in the right-hand side of the graph are IBM and General Electric (GE), which differ from the other companies with their high market values. As can

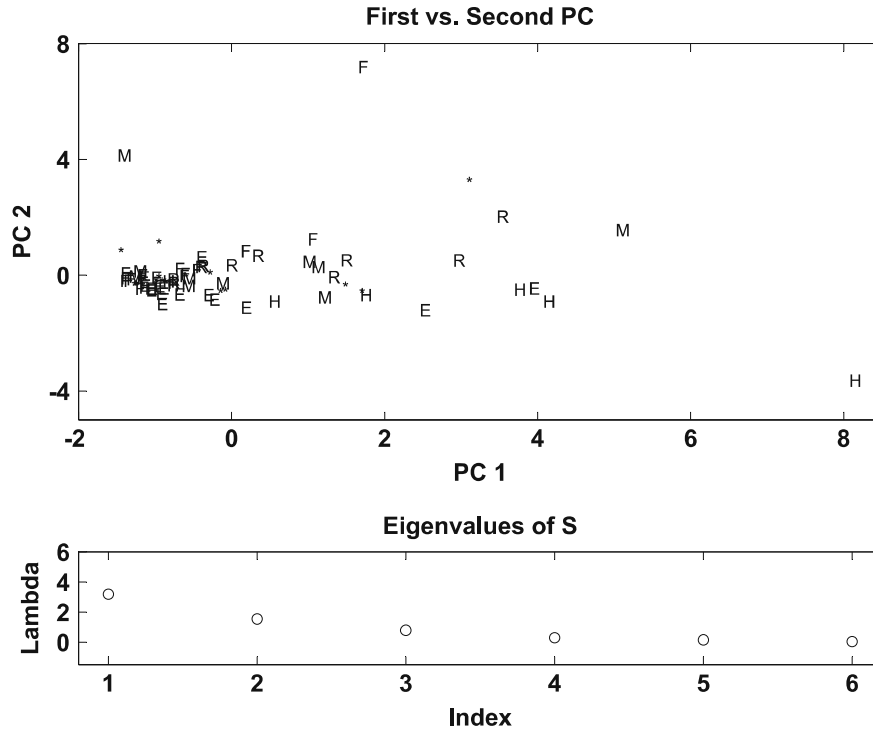


Fig. 11.16 Principal components of the US company data (without IBM and General Electric) 
MVAnpcausco2

be seen in the first column of $\mathcal{G}$, market value has the largest weight in the first PC, adding to the isolation of these two companies. If IBM and GE were to be excluded from the data set, a completely different picture would emerge, as shown in Fig. 11.16. In this case the vector of eigenvalues becomes

$$\ell = (3.191, 1.535, 0.791, 0.292, 0.149, 0.041)^\top,$$

and the corresponding matrix of eigenvectors is

$$\mathcal{G} = \begin{pmatrix} 0.263 & -0.408 & -0.800 & -0.067 & 0.333 & 0.099 \\ 0.438 & -0.407 & 0.162 & -0.509 & -0.441 & -0.403 \\ 0.500 & -0.003 & -0.035 & 0.801 & -0.264 & -0.190 \\ 0.331 & 0.623 & -0.080 & -0.192 & 0.426 & -0.526 \\ 0.443 & 0.450 & -0.123 & -0.238 & -0.335 & 0.646 \\ 0.427 & -0.277 & 0.558 & 0.021 & 0.575 & 0.313 \end{pmatrix}.$$

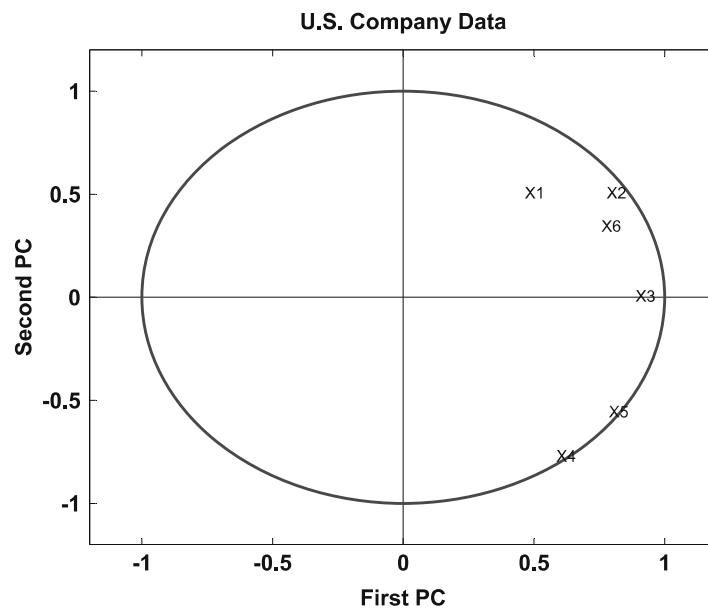
The percentage of variation explained by each component is given in Table 11.9. The first two components explain almost 79 % of the variance. The interpretation of the factors (the axes of Fig. 11.16) is given in the table of correlations (Table 11.10). The first two columns of this table are plotted in Fig. 11.17.

Table 11.9 Eigenvalues and proportions of explained variance

ℓ_j	Proportion of variance	Cumulated proportion
3.191	0.532	0.532
1.535	0.256	0.788
0.791	0.132	0.920
0.292	0.049	0.968
0.149	0.025	0.993
0.041	0.007	1.000

Table 11.10 Correlations with PCs

	$r_{X_i Z_1}$	$r_{X_i Z_2}$	$r_{X_i Z_1}^2 + r_{X_i Z_2}^2$
X_1 : assets	0.47	-0.510	0.48
X_2 : sales	0.78	-0.500	0.87
X_3 : market value	0.89	-0.003	0.80
X_4 : profits	0.59	0.770	0.95
X_5 : cash flow	0.79	0.560	0.94
X_6 : employees	0.76	-0.340	0.70

**Fig. 11.17** The correlation of the original variables with the PCs  MVAnpcausco2i

From Fig. 11.17 (and Table 11.10) it appears that the first factor is a “size effect”, it is positively correlated with all the variables describing the size of the activity of the companies. It is also a measure of the economic strength of the firms. The second factor describes the “shape” of the companies (“profit-cash flow” vs. “assets-sales” factor), which is more difficult to interpret from an economic point of view.

Example 11.10 Volle (1985) analyses data on 28 individuals (Table 22.14). For each individual, the time spent (in hours) on 10 different activities has been recorded over 100 days, as well as informative statistics such as the individual's sex, country of residence, professional activity and matrimonial status. The results of a NPCA are given below.

The eigenvalues of the correlation matrix are given in Table 11.11. Note that the last eigenvalue is exactly zero since the correlation matrix is singular (the sum of all the variables is always equal to $2,400 = 24 \times 100$). The results of the 4 first PCs are given in Tables 11.12 and 11.13.

From these tables (and Figs. 11.18 and 11.19), it appears that the professional and household activities are strongly contrasted in the first factor. Indeed on the horizontal axis of Fig. 11.18 it can be seen that all the active men are on the right and all the inactive women are on the left. Active women and/or single women are in between. The second factor contrasts meal/sleeping vs. toilet/shopping (note the high correlation between meal and sleeping). Along the vertical axis of Fig. 11.18 we see near the bottom of the graph the people from Western-European countries, who spend more time on meals and sleeping than people from the US (who can be found close to the top of the graph). The other categories are in between.

Table 11.11 Eigenvalues of correlation matrix for the time budget data

ℓ_j	Proportion of variance	Cumulated proportion
4.59	0.459	0.460
2.12	0.212	0.670
1.32	0.132	0.800
1.20	0.120	0.920
0.47	0.047	0.970
0.20	0.020	0.990
0.05	0.005	0.990
0.04	0.004	0.999
0.02	0.002	1.000
0.00	0.000	1.000

Table 11.12 Correlation of variables with PCs

		$r_{X_i W_1}$	$r_{X_i W_2}$	$r_{X_i W_3}$	$r_{X_i W_4}$
X_1 :	prof	0.9772	-0.1210	-0.0846	0.0669
X_2 :	tran	0.9798	0.0581	-0.0084	0.4555
X_3 :	hous	-0.8999	0.0227	0.3624	0.2142
X_4 :	kids	-0.8721	0.1786	0.0837	0.2944
X_5 :	shop	-0.5636	0.7606	-0.0046	-0.1210
X_6 :	pers	-0.0795	0.8181	-0.3022	-0.0636
X_7 :	eati	-0.5883	-0.6694	-0.4263	0.0141
X_8 :	slee	-0.6442	-0.5693	-0.1908	-0.3125
X_9 :	tele	-0.0994	0.1931	-0.9300	0.1512
X_{10} :	leis	-0.0922	0.1103	0.0302	-0.9574

Table 11.13 PCs for time budget data

	Z_1	Z_2	Z_3	Z_4
maus	0.0633	0.0245	-0.0668	0.0205
waus	0.0061	0.0791	-0.0236	0.0156
wnus	-0.1448	0.0813	-0.0379	-0.0186
mmus	0.0635	0.0105	-0.0673	0.0262
wmus	-0.0934	0.0816	-0.0285	0.0038
msus	0.0537	0.0676	-0.0487	-0.0279
wsus	0.0166	0.1016	-0.0463	-0.0053
mawe	0.0420	-0.0846	-0.0399	-0.0016
wawe	-0.0111	-0.0534	-0.0097	0.0337
wnwe	-0.1544	-0.0583	-0.0318	-0.0051
mmwe	0.0402	-0.0880	-0.0459	0.0054
wmwe	-0.1118	-0.0710	-0.0210	0.0262
mswe	0.0489	-0.0919	-0.0188	-0.0365
wswe	-0.0393	-0.0591	-0.0194	-0.0534
mayo	0.0772	-0.0086	0.0253	-0.0085
wayo	0.0359	0.0064	0.0577	0.0762
wnyo	-0.1263	-0.0135	0.0584	-0.0189
mmyo	0.0793	-0.0076	0.0173	-0.0039
wmyo	-0.0550	-0.0077	0.0579	0.0416
msyo	0.0763	0.0207	0.0575	-0.0778
wsyo	0.0120	0.0149	0.0532	-0.0366
maes	0.0767	-0.0025	0.0047	0.0115
waes	0.0353	0.0209	0.0488	0.0729
wnes	-0.1399	0.0016	0.0240	-0.0348
mmes	0.0742	-0.0061	-0.0152	0.0283
wmes	-0.0175	0.0073	0.0429	0.0719
mses	0.0903	0.0052	0.0379	-0.0701
fses	0.0020	0.0287	0.0358	-0.0346

In Fig. 11.19 the variables television and other leisure activities hardly play any role (look at Table 11.12). The variable television appears in Z_3 (negatively correlated). Table 11.13 shows that this factor contrasts people from Eastern countries and Yugoslavia with men living in the US. The variable other leisure activities is the factor Z_4 . It merely distinguishes between men and women in Eastern countries and in Yugoslavia. These last two factors are orthogonal to the preceding axes and of course their contribution to the total variation is less important.

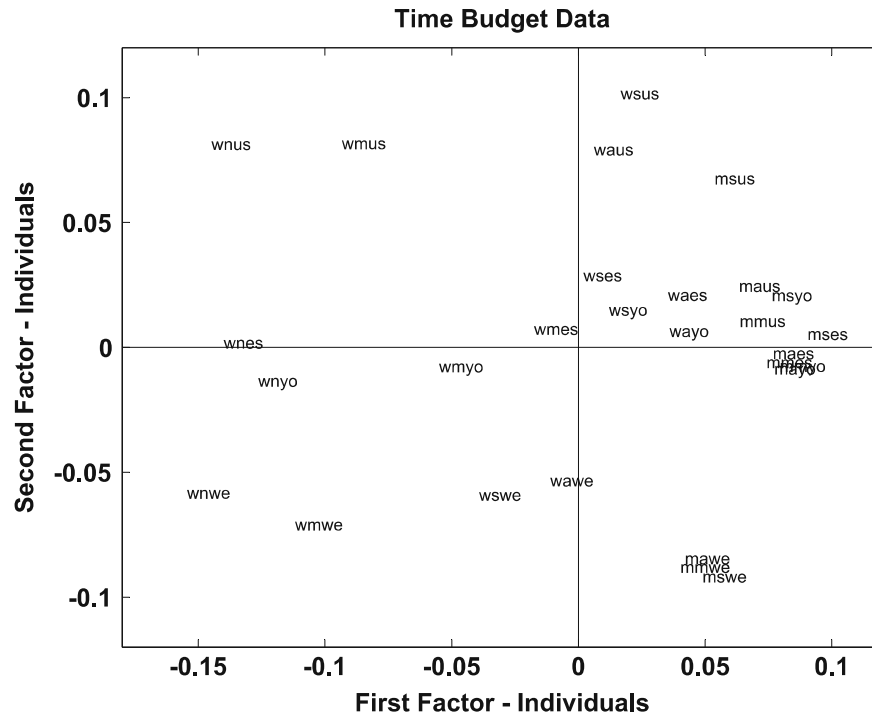


Fig. 11.18 Representation of the individuals  MVAncatime

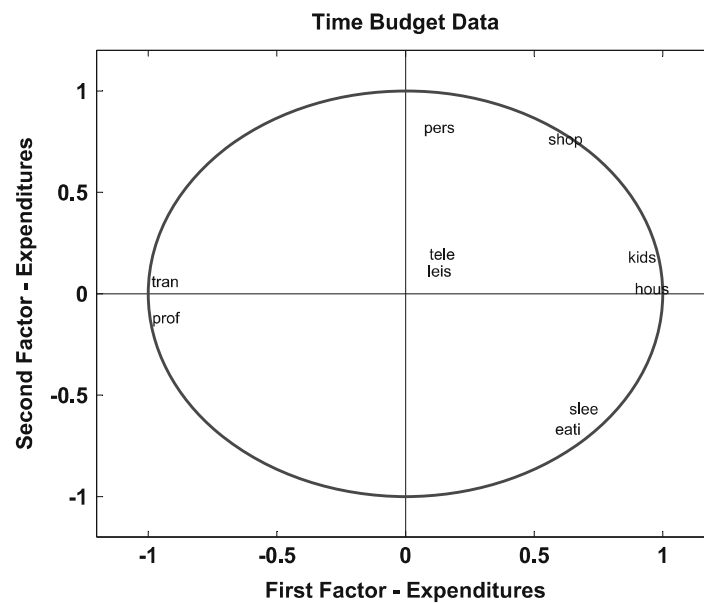


Fig. 11.19 Representation of the variables  MVAncatime

11.10 Exercises

Exercise 11.1 Prove Theorem 11.1. (Hint: use (4.23).)

Exercise 11.2 Interpret the results of the PCA of the US companies. Use the analysis of the bank notes in Sect. 11.3 as a guide. Compare your results with those in Example 11.9.

Exercise 11.3 Test the hypothesis that the proportion of variance explained by the first two PCs for the US companies is $\psi = 0.75$.

Exercise 11.4 Apply the PCA to the car data (Table 22.7). Interpret the first two PCs. Would it be necessary to look at the third PC?

Exercise 11.5 Take the athletic records for 55 countries (Sect. 22.18) and apply the NPCA. Interpret your results.

Exercise 11.6 Apply a PCA to $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, where $\rho > 0$. Now change the scale of X_1 , i.e. consider the covariance of cX_1 and X_2 . How do the PC directions change with the screeplot?

Exercise 11.7 Suppose that we have standardised some data using the Mahalanobis transformation. Would it be reasonable to apply a PCA?

Exercise 11.8 Apply a NPCA to the US CRIME data set (Table 22.10). Interpret the results. Would it be necessary to look at the third PC? Can you see any difference between the four regions? Redo the analysis excluding the variable “area of the state”.

Exercise 11.9 Repeat Exercise 11.8 using the US HEALTH data set (Table 22.16).

Exercise 11.10 Do a NPCA on the GEOPOL data set (see Table 22.15) which compares 41 countries w.r.t. different aspects of their development. Why or why not would a PCA be reasonable here?

Exercise 11.11 Let U be an uniform r.v. on $[0, 1]$. Let $a \in \mathbb{R}^3$ be a vector of constants. Suppose that $X = Ua^\top = (X_1, X_2, X_3)$. What do you expect the NPCs of X to be?

Exercise 11.12 Let U_1 and U_2 be two independent uniform random variables on $[0, 1]$. Suppose that $X = (X_1, X_2, X_3, X_4)^\top$ where $X_1 = U_1$, $X_2 = U_2$, $X_3 = U_1 + U_2$ and $X_4 = U_1 - U_2$. Compute the correlation matrix P of X . How many PCs are of interest? Show that $\gamma_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 1, 0\right)^\top$ and $\gamma_2 = \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0, 1\right)^\top$ are eigenvectors of P corresponding to the non trivial λ 's. Interpret the first two NPCs obtained.

Exercise 11.13 Simulate a sample of size $n = 50$ for the r.v. X in Exercise 11.12 and analyse the results of a NPCA.